JEREMY THUMMEL

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EDUCATION

Ontario Tech University | Oshawa, ON
Bachelor's in Computer Science (Honours) | September 2022 - May 2027
GPA: 4.06 / 4.30 | President's List Member, 2022 - 2026
Relevant Courses: Machine Learning Theory and Applications, Machine Learning II, Natural Language Processing, Data Structures, Analysis and Design of Algorithms, Database Systems & Concepts, Computer Architecture, Linear Algebra, Discrete Mathematics

WORK EXPERIENCE

AI Systems Developer Intern | May 2026 - September 2026
AFM Industries | Whitby, ON

- Developed and deployed a LangGraph AI assistant in Microsoft Teams to answer industrial-belting operations questions and automate employee workflows, achieving an **86% self-service resolution rate**.
- Built a multimodal retrieval-augmented generation (RAG) pipeline using Azure AI Search, SharePoint, PostgreSQL, OCR, and image transcription to retrieve information from technical documents and company data.
- Orchestrated LLM supervisor and evaluation agents within state-based workflows for calculations, quote generation, and approval routing; enforced source-level access permissions through security trimming.
- Analyzed employee feedback using vector similarity and LLM categorization, producing biweekly reports that identified recurring issues and negatively rated interactions.

Cybersecurity and Automation Intern | January 2025 - December 2025
Teknion Furniture Systems | North York, ON

- Automated employee permission updates through Zendesk-triggered Okta Workflows, REST APIs, and custom connectors, **reducing average provisioning time by 38%.**
- Built a configurable JavaScript and Google Apps Script label-printing application supporting multiple manufacturing plants, printers, and label formats.
- Developed SQL access-provisioning schemas and PowerShell validation and reconciliation tools to identify permission discrepancies and test authentication and connectivity across Citrix and internal networks.

LEADERSHIP & ACTIVITIES

Technical Director | October 2024 - April 2025
Google Developer Groups on Campus | Oshawa, ON

- Published and delivered workshops demonstrating NotebookLM research workflows and Cohere language-model integration in Python, helping students apply generative AI and REST API tools in their own projects.

Sports: Intramural Soccer 2026, Intramural Basketball 2024

PROJECTS

Parkinson's Disease Rehabilitation App - Honours Thesis, Ontario Tech University **September 2026 - Current**

- Developing a computer vision model with Python for webcam-based facial rehabilitation for Parkinson's patients, comparing facial movement tracking and reference-based scoring approaches to guide rehabilitation exercise feedback with a TypeScript frontend.

Hack the Heart - Hack the North 2026, University of Waterloo **Live Demo | GitHub Repository | September 2026**

- Co-developed an AI matchmaking app that turns conference-badge contacts into potential connections, using personalized LLM agents to simulate dates and present conversation histories, compatibility scores, and reasons for alignment or friction.
- Designed a multi-agent architecture separating social interpretation from response generation, combining psychology-informed reasoning, user-specific communication profiles, and versioned feedback memory to adapt future interactions.
- Built the Next.js, TypeScript, and MongoDB application with Zod schema validation and source-evidence checks to keep agent responses grounded in user profiles and conversation history.

M.A.G.E. - Masked and Generative Image Editor **GitHub Repository | January 2026 - April 2026**

- Built a Python and Gradio image editor that lets users replace objects and restyle selected regions through natural-language instructions, combining manual brush masks or YOLO instance segmentation with Stable Diffusion inpainting.
- Integrated T5 prompt enhancement and seed-controlled generation to help users explore alternative edits, displaying CLIP image–text similarity and SSIM background-preservation scores to assess results.

Emojiable - Emoji Generation Model **Live Demo | GitHub Repository | September 2024 - December 2024**

- Collaborated in a four-person team to train a PyTorch DCGAN on a multi-platform emoji dataset, using custom generator and discriminator networks, transposed convolutions, batch normalization, and adversarial training to synthesize new emoji designs.
- Integrated the trained generator into a browser app using ONNX Runtime, enabling users to explore seeded variations, customize image appearance, and download PNGs without installing Python or running a backend inference server.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, PowerShell, C++, HTML / CSS, ASM, TypeScript, React, Next.js
AI Tools: Codex, Claude, System Prompt Engineering, MCP and Tool Connection, Custom Plugin Integration, ONNX Runtime
Backend and Automation: REST APIs, WebSockets, Bash Scripting, JSON, Selenium, SMTP, Google Apps Script, LangGraph, PyTorch, Gradio, OpenAI APIs, Cohere APIs, Pydantic, TanStack, Microsoft Azure, Microsoft Foundry, Azure AI Search, Okta Workflows
Databases and Analysis: SQL, PostgreSQL, pgvector, Pandas, NumPy, Jupyter Notebook, Power BI, Supabase, Matplotlib, MongoDB